Conjecture 00000001189: its defined support is zero, not $n^2 - 1$

lizaixi01, with OpenAI Codex assistance

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1 The source's own definition and the necessary contradiction

The source defines minimal support using the minimum of the dimension of the fixed-point space in the natural n-dimensional representation. It then asserts a generating pair with common fixed-space dimension zero, each individual fixed space at most one-dimensional, "giving support $n^2 - 1$." Both languages use this same preceding definition.

These claims are incompatible already for the legitimate parameters $n = 2$, $q = 2$. We prove on the actual group $SL_2(\mathbf{F}_2)$ that a generating pair with all those fixed-space properties exists, and that the actual minimum of fixed dimensions over generating pairs is zero. It cannot be $2^2 - 1 = 3$. More generally, every actual natural fixed subspace has dimension at most two; the minimum over any nonempty family of such image fixed spaces therefore also cannot be three.

This refutation uses the source's stated fixed-dimension quantity. It does not identify it with a different standard support convention, such as a codimension, an adjoint-representation quantity, or an algebraic-group dimension. Nor does it assert that Singer-cycle constructions or the separate generating-pair existence claim fail. The support-value necessary clause already refutes the conjunction. The exact original bilingual source is included in the package.

2 Actual matrices, natural action and fixed spaces

Lean uses Mathlib's actual special linear group $G = SL_2(\mathbf{F}_2)$ and its actual natural action on $V = \mathbf{F}_2^2$. For $A \in G$ define the genuine linear submodule

$$F_A = \ker(A - I) \le V, \qquad F_{A,B} = F_A \cap F_B.$$

The membership theorems identify these kernels with the equations $Av = v$ and $Av = v$, $Bv = v$, respectively. Dimensions are actual module dimensions over the genuine field `ZMod 2`. Since $\dim V = 2$, Lean proves both $\dim F_{A,B} \le 2$ and $\dim V^H \le 2$ for every actual subgroup $H \le G$.

The subgroup generated by a pair is the actual subgroup closure $\langle A, B\rangle$. Its fixed space is defined as the intersection of F_C over all its actual elements C. Lean proves

$$F_{A,B} = V^{\langle A,B\rangle}.$$

Indeed, if v is fixed by A and B, its actual group-action stabilizer contains both generators, hence their subgroup closure. The reverse implication follows because the generators belong to their closure. This connects common fixed spaces to the source's fixed space of the generated natural representation image.

3 A genuine generating pair, with all fixed-space conditions

Take the actual determinant-one transvections

$$U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad L = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \quad \text{over } \mathbf{F}_2.$$

The formal proof uses Mathlib's special-linear transvection generation theorem. In dimension two over $\mathbf{F}_2$, every transvection is either the identity, U, or L: the off-diagonal position has two choices and its coefficient is zero or one. The subgroup containing U and L therefore contains every transvection and, by that genuine generation theorem, all of G. Thus their actual subgroup closure is G.

For $v = (x, y)$ the actual action gives

$$Uv = v \iff y = 0, \qquad Lv = v \iff x = 0.$$

Consequently $F_{U,L} = \{0\}$ and its dimension is zero. Each individual fixed submodule is proper: $(0, 1)$ is not fixed by U, and $(1, 0)$ is not fixed by L. The strict dimension bound for a proper subspace of V proves that both fixed dimensions are at most one. The theorem `actual_pair_satisfies_fixed_conditions` collects all these genuine generating-pair and fixed-space properties.

4 The attained minimum and other image families

The literal pair-based set of admissible dimensions is

$$S = \{\dim F_{A,B} : A, B \in G,\ \langle A, B\rangle = G\}.$$

The verified pair U, L proves $0 \in S$, so this is a nonempty set of actual dimensions. Nonnegativity gives $\mathrm{IsLeast}(S, 0)$. Lean defines the minimum by $\inf S$ and derives its value zero from that attained least-element theorem. No default value of an infimum on an empty set is used.

If the minimum syntax is instead understood as ranging over another nonempty collection of natural representation images, the conclusion still cannot be three. Lean separately quantifies over every actual nonempty set $\mathcal{H}$ of subgroups of G and proves

$$\inf\{\dim V^H : H \in \mathcal{H}\} \leq 2.$$

This uses an actual member of the nonempty collection and the genuine ambient dimension bound. For the full image G itself, Lean also proves $V^G = \{0\}$ using the pair-generation bridge above. Thus neither of these fixed-space interpretations of the source's minimum can yield $n^2 - 1$ at $n = 2$.

5 Verification

Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`

are pinned. The package includes a new root build, transitive axiom audits for every theorem and full official `leanchecker --fresh Main` replay. No `sorry`, `native_decide` or extra axiom is used. OpenAI Codex assisted; a separate supervising Codex agent reviewed source, report and PDF. This is AI review, not human certification. Organizer acceptance is pending.